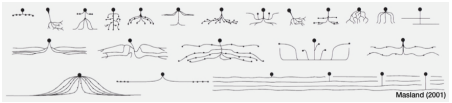
Functional heterogeneity of wide-field amacrine cells with a common molecular marker

Juyoung Kim[1], Masatoshi Inoue[2], Charu Ramakrishnan[2], Karl Deisseroth[2,3,4], Stephen A. Baccus[1]

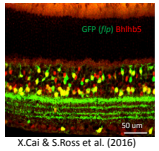
[1]Neurobiology, Stanford University School of Medicine,
[2]Bioengineering, [3]Psychiatry and Behavioral Sciences, [4]Howard Hughes Medical Institute, Stanford University, Stanford, CA

Amacrine cells in retina

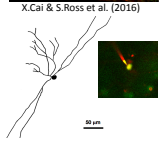


Amacrine cells are the most diverse cells (> 30 types) in retina. They differ in dendritic field sizes (e.g. narrow- or wide-field) and in arborization patterns in IPL (e.g. monostratified or diffuse). Most have unknown function. **Few studies have used optical recording from genetically targeted amacrine cell populations.**

Bhlhb5-flp

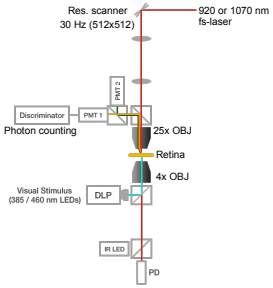


X.Cai & S.Ross et al. (2016)



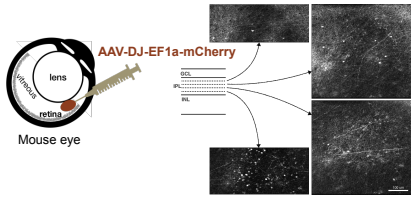
Bhlhb5-flp labels a subset of GABAergic wide-field amacrine cells with long and nearly unbranched processes.
What are the functions of these cells, and how diverse is this population?

2-photon scope



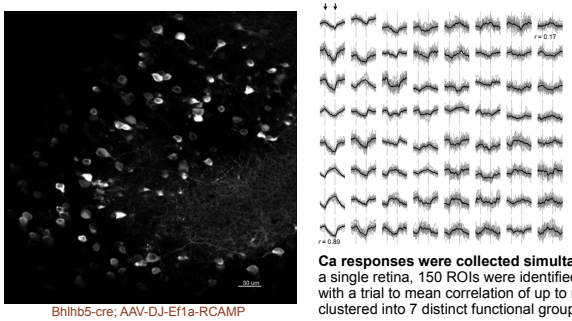
We built a custom 2-photon scope for optically recording visual responses from targeted populations in the retina. **High-speed Ca imaging, red Ca indicator, photon counting and genetic tools for targeting are combined in this study.**

Imaging Bhlhb5-cre: Viral approach

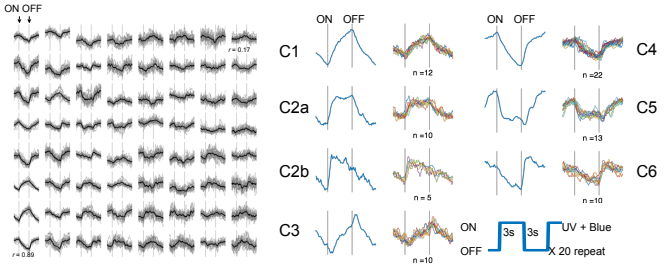


- AAV-mCherry was intravitreally delivered to an adult Bhlhb5-cre mouse retina.
- Amacrine cells with long linear processes were identified from 4 different levels in IPL, consistent with Bhlhb5-expressing amacrine cells in the retina (Feng et al., 2006).
- Cre knock-in mouse line (Ross et al., 2010) was a gift from Sarah Ross (U. Pittsburgh).

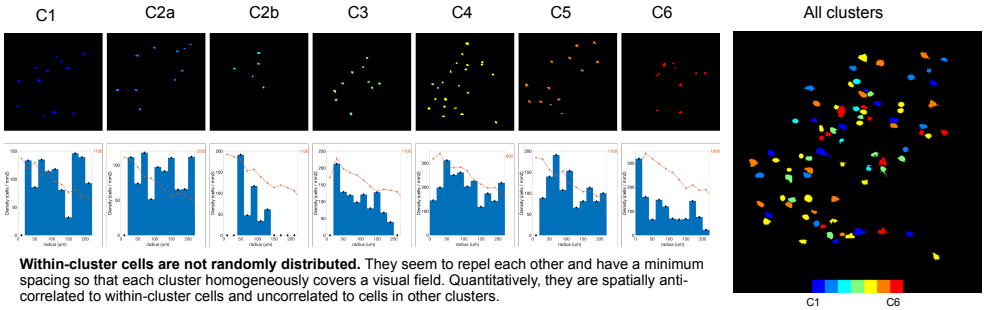
Heterogeneous responses from a common molecular marker



Bhlhb5-cre; AAV-DJ-Ef1a-RCAMP

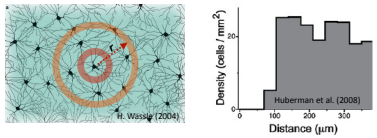


Ca responses were collected simultaneously from a Bhlhb5-targeted amacrine cell population. In a single retina, 150 ROIs were identified. Cells exhibited highly reliable responses to repeated flashes, with a trial to mean correlation of up to nearly 0.9. 82 average responses (r >0.1) were selected and clustered into 7 distinct functional groups.



Within-cluster cells are not randomly distributed. They seem to repel each other and have a minimum spacing so that each cluster homogeneously covers a visual field. Quantitatively, they are spatially anti-correlated to within-cluster cells and uncorrelated to cells in other clusters.

Density Recovery Profile

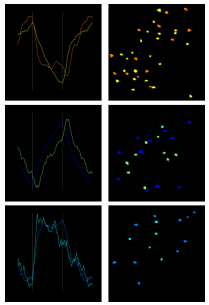


Density Recovery Profile (r) :

$$= \sum \frac{\text{Number of cells in the ring } (r \sim r+dr)}{\text{Available area } (r \sim r+dr)}$$

(Rodieck, 1991)

Cluster comparisons

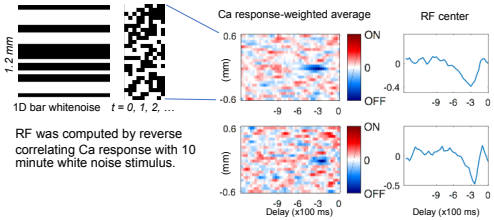


C4 vs C5: The most numerous groups. Both are OFF types. Cells in the two clusters were often closer than the within cluster minimum distance.

C1 vs C3: Functional responses look similar, although a 0.5 sec delay is clearly visible. Moreover, each cluster covers the retina independently.

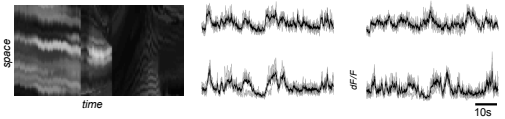
C2a vs C2b: Cells in C2a are located at the center whereas C2b cells are located in the periphery. The position of the 600 μm flash stimulus could explain the difference in the response, suggesting these clusters may be from the same cell type.

Receptive field mapping by red Ca imaging



RF was computed by reverse correlating Ca response with 10 minute white noise stimulus.

Reliable responses to naturalistic movies



Example of 1D naturalistic movie. One row of pixels from real natural movies + fixational / saccadic eye movements.

Optical responses to 1D naturalistic movies using . hSyn-GCaMP6f. One min. stimulus repeated 5 times. Trial to mean correlation for above cells > 0.5.

Summary

1. Video rate visual responses of a population of Bhlhb5 wide-field amacrine cells were optically measured under various stimuli.
2. Cells showed up to seven distinct response types to a simple flash. Clustered groups of cells exhibit self-avoidance, suggesting that they likely comprise multiple individual cell types.
3. Ca responses enable computation of spatiotemporal receptive fields, and the high reliability to naturalistic stimuli indicate complex modeling is possible from optical population recordings.
4. Bhlhb5 amacrine cells consist of multiple cell types with distinct responses, whose spatiotemporal neural code can be analyzed using optical recording.

Acknowledgement

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